The Castle Crown Caper

Whodunnit? · Polynomials & Factoring · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Polynomials & Factoring	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Factoring trinomials, differences of squares, GCFs, and multiplying binomials.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Wizard Orrin took it, in the Tower Library, using this method: Smuggled in a laundry basket.

The reveal (read aloud): During the trumpet fanfare, **Wizard Orrin** slipped into the **Tower Library**, tucked the Royal Crown into a **laundry basket**, and wheeled it right past the guards. Every alibi and method you cleared led straight to him. Long live the detective!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Lady Wren · Place: Throne Room
2	Method: Hidden in a treasure chest
3	Suspect: Sir Bram · Place: Drawbridge
4	Method: Tucked inside a lute
5	Suspect: Jester Pip · Place: Dungeon
6	Method: Lowered from a window
7	Suspect: Chef Margaux · Place: Royal Kitchen
8	Method: Hidden in a suit of armor
9	Suspect: Bard Elowen · Place: Rose Garden
10	Method: Carried in a magic hat

Answer Key

The Castle Crown Caper · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Factor: $x^2 + 5x + 6$	**B.** $(x + 2)(x + 3)$	find two numbers: product 6, sum 5 → 2 and 3 $(x + 2)(x + 3)$	Lady Wren was rehearsing the coronation speech in the Throne Room before a dozen courtiers.
2	Factor: $x^2 - 5x + 6$	**D.** $(x - 2)(x - 3)$	product 6, sum -5 → -2 and -3 $(x - 2)(x - 3)$	The treasure chest was opened in front of the guards. It was full of old coins.
3	Factor: $x^2 + x - 6$	**B.** $(x - 2)(x + 3)$	product -6, sum 1 → 3 and -2 $(x - 2)(x + 3)$	Sir Bram was guarding the Drawbridge. Every knight on duty saw him.
4	Factor: $x^2 - x - 6$	**D.** $(x + 2)(x - 3)$	product -6, sum -1 → -3 and 2 $(x + 2)(x - 3)$	The lute’s hollow body was empty. Only a mouse lived inside.
5	Factor: $x^2 - 9$	**A.** $(x - 3)(x + 3)$	difference of squares: $a^2 - b^2 = (a - b)(a + b)$ $(x - 3)(x + 3)$	Jester Pip was juggling for the prisoners in the Dungeon. The guard laughed the whole time.
6	Factor completely: $2x^2 + 8x$	**A.** $2x(x + 4)$	GCF = $2x$ $2x(x + 4)$	Rope marks on the window are old and dusty. Nothing was lowered recently.
7	Factor: $2x^2 + 7x + 3$	**C.** $(2x + 1)(x + 3)$	multiply $2 \cdot 3 = 6$; find two numbers with sum 7: 1 and 6 split the middle: $2x^2 + x + 6x + 3 = x(2x + 1) + 3(2x + 1)$ $(2x + 1)(x + 3)$	Chef Margaux was frosting the coronation cake in the Royal Kitchen.
8	Factor: $x^2 - 6x + 9$	**A.** $(x - 3)^2$	perfect square trinomial: $(x - 3)(x - 3)$ $(x - 3)^2$	The suits of armor were all checked. Each was empty.
9	Multiply: $(x + 2)(x - 5)$	**C.** $x^2 - 3x - 10$	$x^2 - 5x + 2x - 10$ $x^2 - 3x - 10$	Bard Elowen was singing under the roses in the Rose Garden. Three ladies listened.
10	Factor completely: $3x^2 - 12$	**D.** $3(x - 2)(x + 2)$	GCF = 3: $3(x^2 - 4)$ $3(x - 2)(x + 2)$	The magic hat is a flat, folded hat. It can’t hide a crown.

Wrong-Answer Map

The Castle Crown Caper · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. $(x - 2)(x - 3)$	sign error	Sir Bram was polishing his sword in the Tower Library. The librarian shushed him twice.
1	C. $(x + 5)(x + 6)$	used b and c as the numbers	Every laundry basket in the castle was dumped out. Only socks and tablecloths were found.
1	D. $(x + 1)(x + 6)$	product 6 but sum 7	Wizard Orrin was casting a growth spell on the Rose Garden. The petals are still glittering.
2	A. $(x - 5)(x - 6)$	used b and c as the numbers	The laundry maid says her basket never left the washhouse. Its wicker lid is still tied shut.
2	B. $(x - 1)(x - 6)$	product 6 but sum -7	Jester Pip was juggling scrolls in the Tower Library. The librarian wrote him a scolding note with the time on it.
2	C. $(x + 2)(x + 3)$	ignored the sign of b	Wizard Orrin was presenting a crystal ball in the Throne Room. The guards watched the whole thing.
3	A. $(x + 2)(x - 3)$	signs swapped	Wizard Orrin was testing the Drawbridge chains with a sparkly spell. The gatekeeper took notes.
3	C. $(x - 1)(x + 6)$	product -6 but sum 5	Lady Wren was reading old treaties in the Tower Library. Her name is on the sign-in scroll.
3	D. $(x + 1)(x - 6)$	product -6 but sum -5	The big laundry basket was weighed at sunset. It was light as a feather.
4	A. $(x + 1)(x - 6)$	product -6 but sum -5	Chef Margaux was hunting for a lost recipe in the Tower Library. The dusty shelves show her fingerprints.
4	B. $(x - 2)(x + 3)$	signs swapped	Wizard Orrin was stirring a bubbling potion in the Royal Kitchen. The kitchen helpers all saw the purple smoke.
4	C. $(x - 1)(x + 6)$	product -6 but sum 5	The guards looked in every laundry basket at the gate. Each held only clean linens.
5	B. $(x - 9)(x + 9)$	used 9 instead of $\sqrt{9}$	The laundry baskets were all stacked empty in the washhouse by noon.
5	C. $(x - 3)^2$	treated difference of squares as a perfect square	Wizard Orrin was turning rats into mice in the Dungeon. The jailer is still shaking his head.
5	D. $(x + 3)^2$	perfect square with wrong sign	Bard Elowen was tuning a lute in the Tower Library. The echoing notes drew a small crowd.
6	B. $2(x + 4)$	dropped the x factor	The castle laundry list matches the number of baskets exactly. None went missing.
6	C. $2x(x + 8)$	did not divide 8 by 2	Wizard Orrin was chatting with a talking toad in the Rose Garden. Two gardeners overheard.
6	D. $2x(x + 2)$	divided 8 by 2 twice	Jester Pip was hiding a rubber chicken among the books in the Tower Library. The librarian caught him red-handed.
7	A. $(2x - 1)(x - 3)$	sign error	Sir Bram was studying battle maps in the Tower Library. The map table has his muddy bootprints.
7	B. $(2x + 1)(x - 3)$	sign error	Not one laundry basket was seen leaving the castle gates today. The gate log is clean.
7	D. $(2x + 3)(x + 1)$	swapped the constants	Wizard Orrin was practicing a fanfare spell in the Throne Room. The court herald counted every note.
8	B. $(x - 6)^2$	used b as the number	The washerwomen counted their baskets twice. Each one went home with its owner.

8	C. $(x + 3)^2$	sign error	Wizard Orrin was turning pumpkins into coaches in the Royal Kitchen. The cook's apprentices cheered.
8	D. $(x - 3)(x + 3)$	treated as difference of squares	Bard Elowen was composing a ballad in the Tower Library. Her inkwell is still wet.
9	A. $x^2 + 3x - 10$	sign error on middle term	Wizard Orrin was feeding the moat monster from the Drawbridge. A passing farmer waved hello.
9	B. $x^2 - 10$	dropped the middle terms	Lady Wren was signing royal letters in the Tower Library. Her wax seal is still warm.
9	D. $x^2 - 3x + 10$	sign error on constant	The laundry basket in the corridor holds only wet capes. The guard checked twice.
10	A. $3(x^2 - 4)$	did not factor completely	A tag on the laundry basket shows it was locked in the storeroom all night.
10	B. $3(x - 2)^2$	treated as perfect square	Chef Margaux was borrowing a cookbook from the Tower Library. The checkout card has the time.
10	C. $3(x - 4)(x + 4)$	used 4 instead of $\sqrt{4}$	Wizard Orrin was chatting with a ghost in the Dungeon. The guard heard the whole conversation.